



PRAYAS

2.0 2027

Physical Chemistry

Lecture – 06

Some Basic Concepts of Chemistry

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Topics to be covered



- 1 Volume contraction
- 2 %age yield & %age purity, limiting reagent
- 3 Questions, Home work from modules
- 4 ★★★★★ *quick*



Rules to Attend Class

- 1. Always sit in a peaceful environment with headphone and be ready with your copy and pen.**
- 2. Never ever attend a class from in between or don't join a live class in the middle of the chapter.**
- 3. Make sure to revise the last class before attending the next class & always complete your home work.**
- 4. Never ever engage in chat whether live or recorded on the topic which is not being discussed in current class.**



Rules to Attend Class

- ✓ 5. Try to make maximum notes during the class if something is left then u can use the notes pdf after the class to complete the remaining class.
- ✓ 6. Always ask your doubts in doubt section to get answer from faculty. Before asking any doubt please check whether same doubt has been asked by someone or not.
7. It does not matter whatever situation you are in **NEVER EVER CREATE A BACKLOG.**



There is one big flaw in your Preparation that's name is Backlog ? What do we say to Backlog ?



NOT TODAY !!!

Question (JEE Main 22nd Jan. 2026, Shift-I)

In the reaction, $2\text{Al(s)} + 6\text{HCl(aq)} \longrightarrow 2\text{Al}^{3+}(\text{aq}) + 6\text{Cl}^{-}(\text{aq}) + 3\text{H}_2(\text{g})$

- A** ☒ 12 L HCl(aq) is consumed for every 6L H₂(g) produced. $\frac{3}{2} \text{ mol}$
 $\frac{3}{2} \times 22.4 \text{ L} = 33.6 \text{ L}$
STP
- B** ☒ 11.2 L H₂(g) at STP is produced for every mole of HCl consumed. $\frac{3}{6} = \frac{1}{2} \text{ mole}$
 $\frac{1}{2} \times 22.4 = 11.2 \text{ L}$
- C** ☒ 67.2 L H₂(g) at STP is produced for every mole of Al that reacts.
- D** ☒ 33.6 L H₂(g) is produced regardless of temperature and pressure for every mole of Al that reacts.

Question [JEE Mains, 31 Jan 2024 (Shift-II)]

A sample of CaCO_3 and MgCO_3 weighed 2.21 g is ignited to constant weight of 1.152 g. The composition of mixture is:

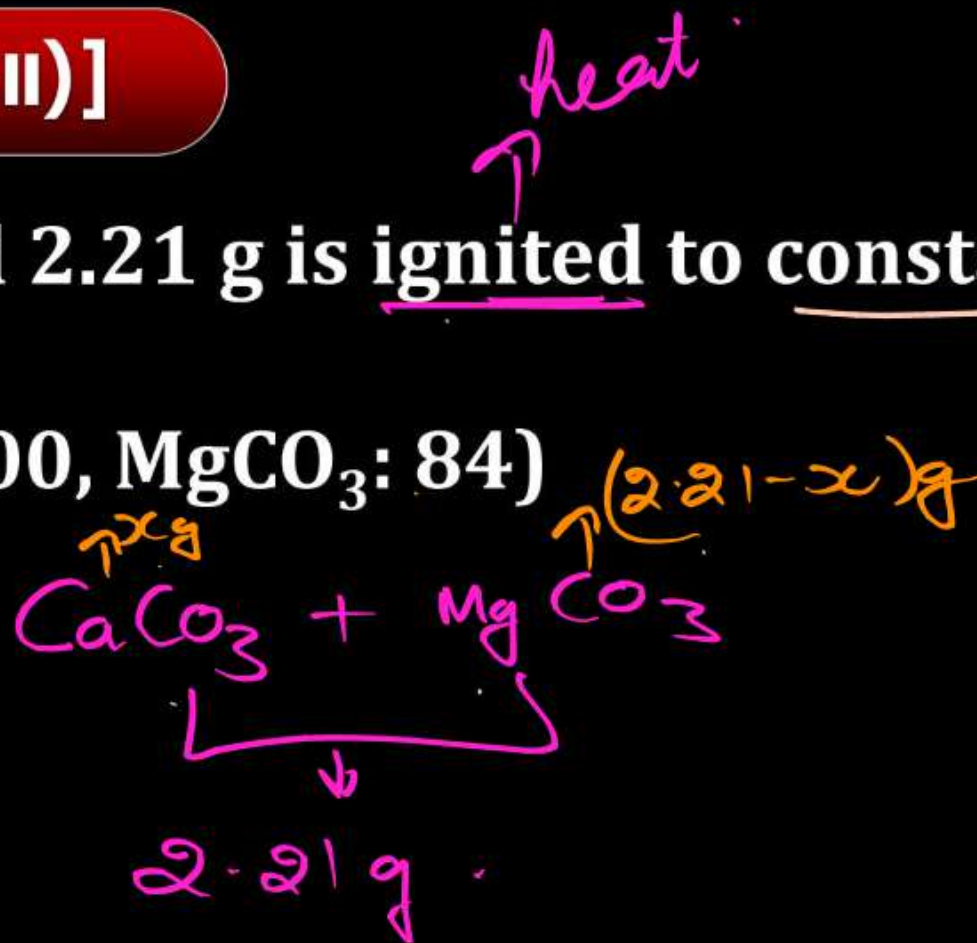
(Given molar mass in g mol^{-1} CaCO_3 : 100, MgCO_3 : 84)

☒ A 1.187 g CaCO_3 + 1.023 g MgCO_3

☐ B ~~X~~ 1.023 g CaCO_3 + 1.023 g MgCO_3

☐ C ~~X~~ 1.187 g CaCO_3 + 1.187 g MgCO_3

☐ D 1.023 g CaCO_3 + 1.187 g MgCO_3





$$\left(\frac{2.21-x}{84} \right) \longrightarrow \left(\frac{2.21-x}{84} \right)$$

$$\frac{(2.21-x) \times 40}{84}$$

$$\frac{56x}{100} + \frac{(2.21-x)40}{84} = 1.152$$

$$\frac{56x \times 84 + 4000(2.21-x)}{8400} = 1.152$$

$$4704x + 8840 - 4000x = 9676.8$$

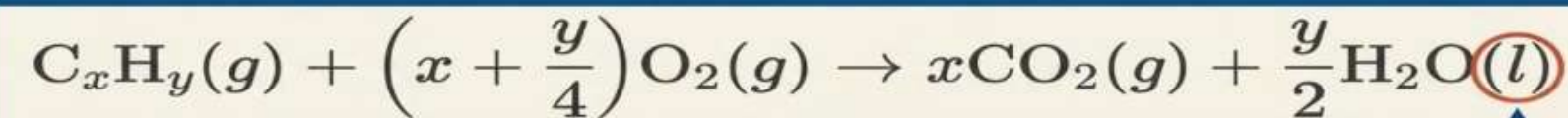
$$704x = 836.8$$

$$x = 1.18 \text{ g}$$



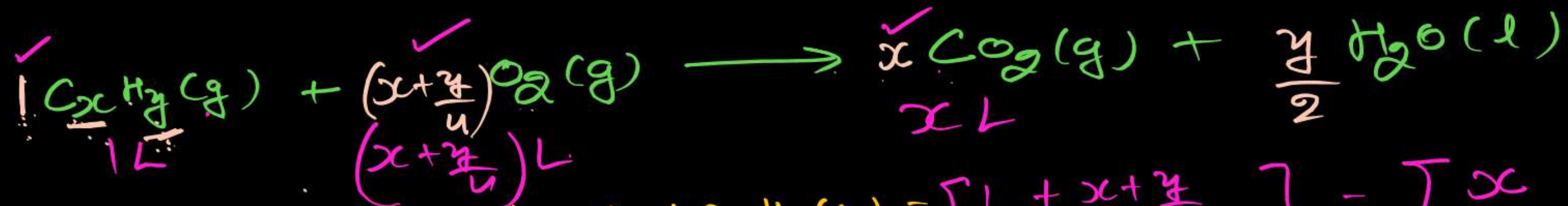
Balancing of Combustion of Organic Compounds

The Organic Combustion Matrix

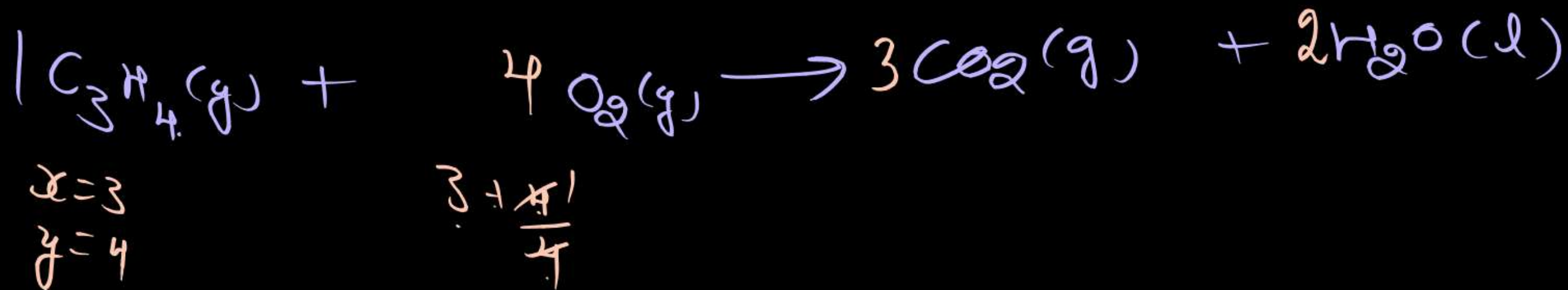
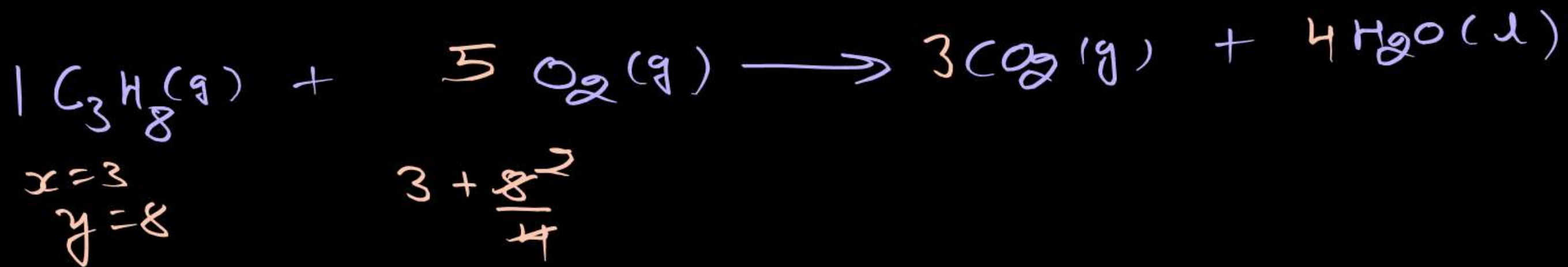


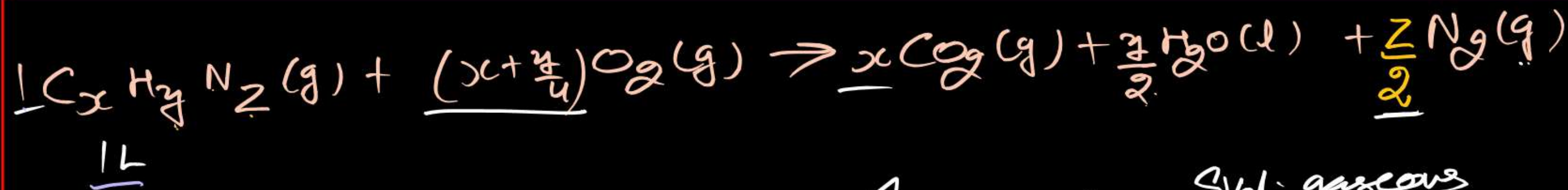
Crucial Detail: At standard conditions, H₂O is a liquid (l). It does not count toward gaseous volume calculations!

#



$$\text{Volume Contraction for } \underline{1L} \text{ of } C_x H_y (g) = \left[1 + x + \frac{y}{4}\right] - [x] = \left(1 + \frac{y}{4}\right)L$$





Volume Contraction for 1 L of $C_x H_y N_z (g)$ = $\sum \text{Vol. gaseous reactant} - \sum \text{Vol. gaseous product}$

$$= \left[1 + x + \frac{y}{4} \right] - \left[x + \frac{z}{2} \right] L$$

$$= \left[1 + \frac{y}{4} - \frac{z}{2} \right] L$$



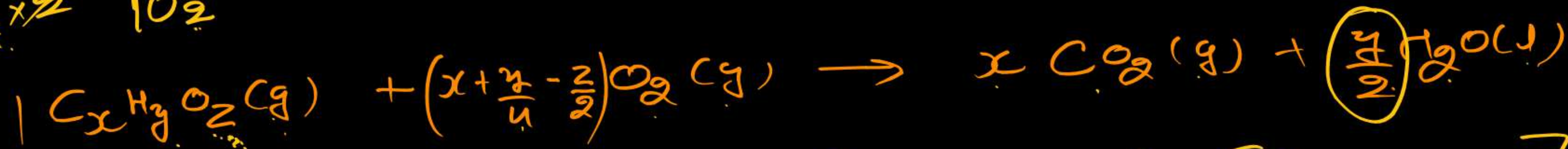
$$\begin{aligned} x &= 3 \\ y &= 9 \\ z &= 1 \end{aligned}$$

$$\frac{3 + \frac{9}{4}}{4} = \frac{21}{4}$$

Vol. Cont. for 1 L of $C_3 H_9 N (g)$

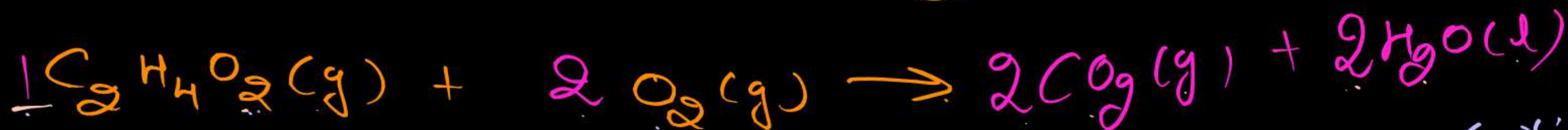
$$= 1 + \frac{9}{4} - \frac{1}{2} = \frac{4 + 9 - 2}{4} = \frac{11}{4} L$$

$\frac{2}{9} \times 2$ 102



Volume Contraction for 1 L of $C_x H_y O_z (g) = \left[1 + x + \frac{y}{4} - \frac{z}{2}\right] - [x]$ L

$$= \left[1 + \frac{y}{4} - \frac{z}{2}\right] L$$



$x=2$
 $y=4$
 $z=2$

$$2 + \frac{4}{4} - \frac{2}{2}$$

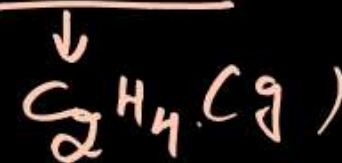
$40x + 20x = 60x$ atoms

$20x + 40x = 60x$

Vol. Contraction for 1 L of $C_2 H_4 O_2 (g) = [1 + 2] - [2]$ L = 1 L

Question

The volume of oxygen required for complete combustion of 20 ml of ethene is



A 30 ml



B 60 ml

$x=2$
 $y=4$

$2 + \frac{4}{4}$

C 40 ml

1 ml $\longrightarrow$ 3 ml

D 50 ml

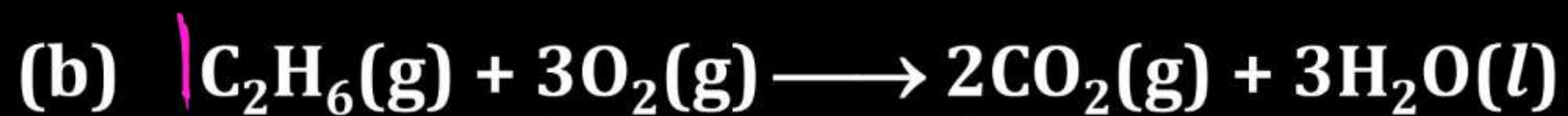
20 ml $\longrightarrow 3 \times 20 = 60 \text{ ml}$

Question

Find the volume contraction of following reaction:



Vol. Contraction for 1 L of $\text{C}_3\text{H}_8(\text{g}) = 6 - 3 = 3 \text{ L}$



Vol. Cont. = $[1 + 3] - 2 = 2 \text{ L}$



%age Yield

$$\frac{8}{80} \times 100 = 10\%$$



#1 T

$$\% \text{ age yield} = \frac{\text{Actual yield} \times 100}{\text{Theoretical yield}}$$

↓
from S.C.

THE REALITY OF PRODUCTS: % YIELD



Theoretical Yield
(Pen & Paper calculation)

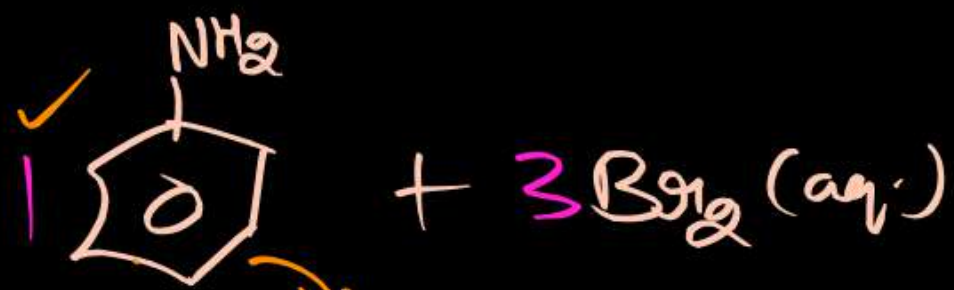


Actual Yield
(Lab Reality - Always less)

$$\% \text{ Yield} = (\text{Actual Yield} / \text{Theoretical Yield}) \times 100.$$

Question [JEE Mains 5 April, 2024 (Shift-I)]

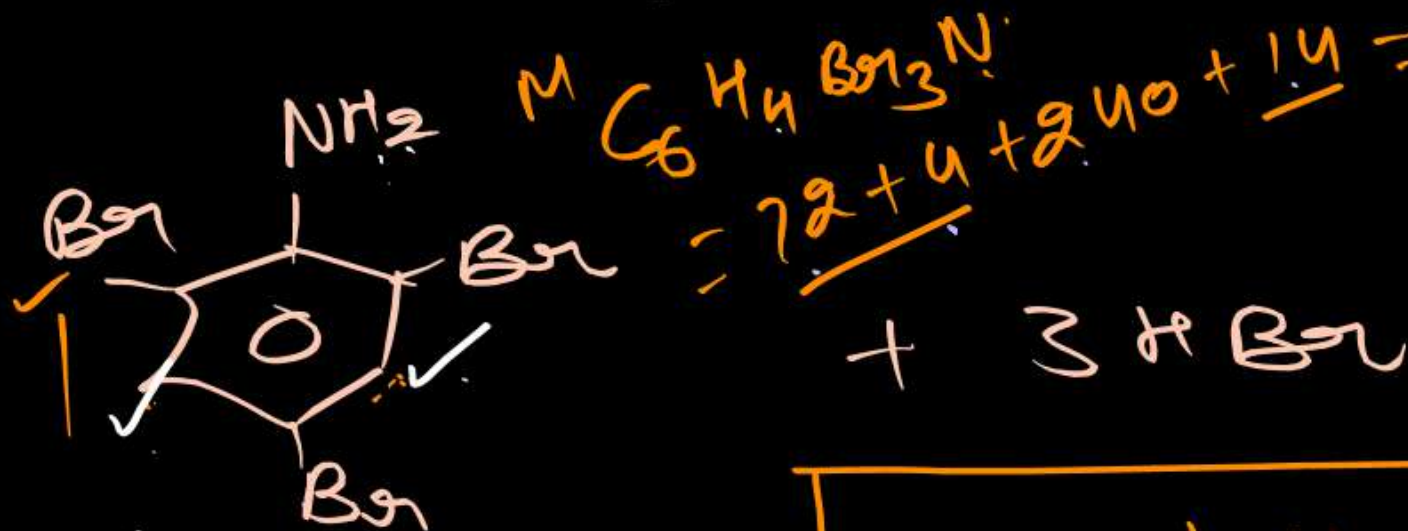
9.3 g of pure aniline is treated with bromine water at room temperature to give a white precipitate of the product 'P'. The mass of product 'P' obtained is 26.4 g. The percentage yield is _____ %



Aniline $M_{C_6H_7N} = 93 \text{ g}$
9.3 g

$$n_g = \frac{9.3}{93} = 0.1 \text{ mol}$$

$$\% \text{ yield} = \frac{26.4}{\text{Theo. yield}} \times 100 = \frac{26.4}{33} \times 100 = 80\%$$



White p.p.t ↓

(P) = 26.4 g

$M_{C_6H_4Br_3N} = 72 + 4 + 240 + 14 = 330 \text{ g/mol}$

$$n_f = \frac{1}{1} \times 0.1 = 0.1 \text{ mol}$$

$$w_{\text{find}} = 0.1 \times 330 = 33 \text{ g}$$

Question



10 g of CaCO_3 on heating gives 5 g of the residue (as CaO). The percent yield of the reaction is approximately:

go back in
page
↑ solid
↓ liquid

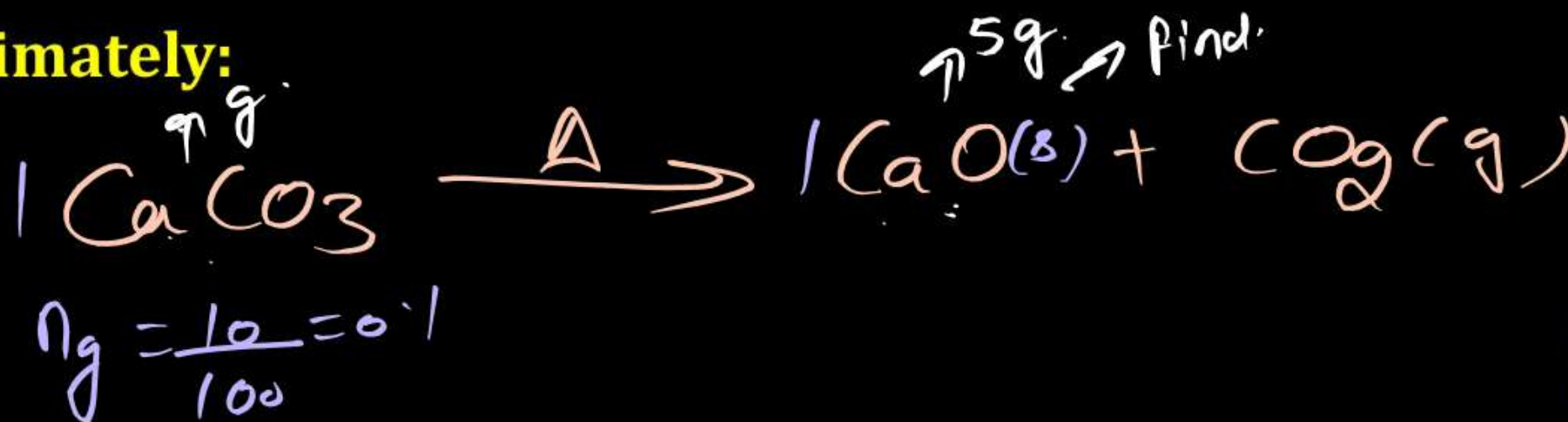
$$M_{\text{CaCO}_3} = 100 \text{ g/mol.}$$
$$M_{\text{CaO}} = 56 \text{ g/mol.}$$

A 50%

B 72%

C 89%

D 100%



$$\% \text{ yield} = \frac{5}{\text{Th. yield}} \times 100$$
$$= \frac{5 \times 1000}{56} = \frac{5000}{56} = 89.2\%$$

$$n_{\text{find}} = \frac{1}{1} \times 0.1 = 0.1$$

$$n_{\text{CaO}} = 0.1 \text{ mol.}$$

$$w_{\text{CaO}} = 0.1 \times 56 = 5.6 \text{ g.}$$



%age Purity



#T
①

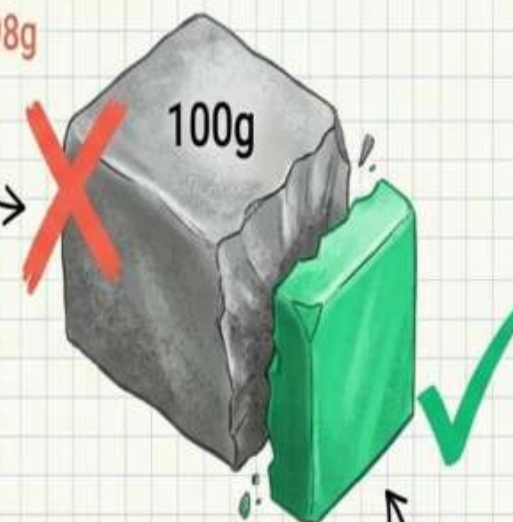
$$\% \text{ Purity} = \frac{\text{mass of pure substance}}{\text{Total mass}} \times 100$$

② in S.C. $\rightarrow$ mass of pure substance

Concept 1: Percentage Purity (The Reality of Reactants)

Reactants are rarely 100% pure in the real world. Only the pure fraction participates in the chemical reaction.

Bhoosa (Impurity) = 98g



The Golden Rule

All calculations must be done according to the mass of the pure substance.

$$\% \text{ Purity} = \frac{\text{Mass of Pure Substance}}{\text{Total Mass of Sample}} \times 100$$

Question

33.6 g of an impure sample of sodium bicarbonate when heated strongly gave 4.4 g of CO_2 . The percentage purity of NaHCO_3 would be:

- ☐ A 25%
- ☒ B 50%
- ☐ C 75%
- ☐ D 100%



$$M_{\text{CO}_2} = 44 \text{ g/mol}$$
$$M_{\text{NaHCO}_3} = 84 \text{ g/mol}$$



$$\text{Total mass NaHCO}_3 = 33.6 \text{ g}$$

$$\% \text{ purity NaHCO}_3 = \frac{\text{Pure NaHCO}_3}{\text{Total NaHCO}_3} \times 100$$



$$n_f = \frac{2}{1} \times 0.1 = 0.2 \text{ mol}$$

$$w_{\text{NaHCO}_3} = 0.2 \times 84$$
$$= 16.8 \text{ g}$$

$$n_g = \frac{4.4}{44} = 0.1 \text{ mol}$$

$$\% \text{ age purity} = \frac{16.8}{33.62} \times 100$$

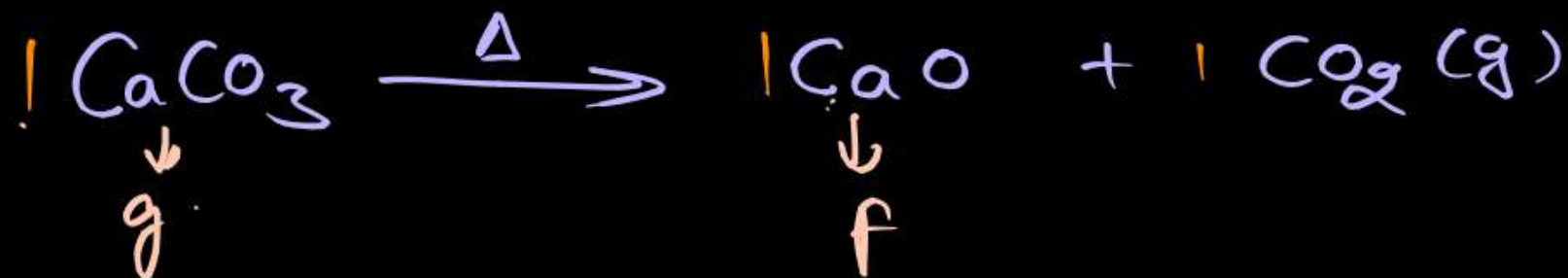
$$= 50.1\%$$

Question (NCERT: PL-20 | NV, JEE Main April 04, 2025 (II))

$$M_{\text{CaCO}_3} = 100 \text{ g/mol} \quad M_{\text{CaO}} = 56 \text{ g/mol}$$

The amount of calcium oxide produced on heating 150 kg limestone (75% pure) is 63 kg. (Nearest integer)

Given : Molar mass (in g mol^{-1}) of Ca - 40, O - 16, C - 12.



$$n_{\text{g}} = \frac{112500}{100} = 1125$$

$$n_{\text{CaO}} = \frac{1}{1} \times 1125$$

$$w_{\text{CaO}} = 1125 \times 56 \text{ g} = \frac{1125 \times 56}{1000} = 63 \text{ kg}$$



$$\frac{75}{100} \times 150 = \frac{225}{2}$$

$$\begin{aligned} w_{\text{pure}} &= 112.5 \text{ kg} \\ \text{CaCO}_3 &= \frac{1125}{10} \times 1000 \\ &= 112500 \text{ g} \end{aligned}$$

Question

$M_{\text{NaCl}} = 58.5 \text{ g/mol}$
 $M_{\text{AgCl}} = 143.5 \text{ g/mol}$ (JEE-Advance)



An sample of NaCl with 50% purity is added into aqueous solution having excess AgNO_3 . AgCl precipitate is filtered, washed and dried weighing 2.87 gm. If the reaction yield is 20%, then find the amount of sample.

A 23.4 gm

B 11.7 gm

C 20 gm

D 12.8 gm

50% Pure

NaCl

↓

find

excess
+ $\text{AgNO}_3(\text{aq.})$

→

given
AgCl ↓

2.87 g

+ $\text{NaNO}_3(\text{aq.})$

100% yield = 20%

$$\% \text{ yield} = \frac{w_{\text{AgCl}}}{w_{\text{AgCl}}^{\text{S.C.}}} \times 100$$

$$2\% = \frac{2.87}{w_{\text{AgCl}}} \times 100$$

$$w_{\text{AgCl}} = 14.35 \text{ g}$$

↓
given.

$$n_{\text{AgCl}} = \frac{14.35}{143.5} = 0.1 \text{ mol}$$

$$n_{\text{NaCl}} = \frac{1}{1} \times 0.1 = 0.1 \text{ mol}$$

$$w_{\text{NaCl}} = 0.1 \times 58.5 = 5.85 \text{ g}$$

$$\% \text{ Purity} = \frac{5.85}{\text{Total mass NaCl}} \times 100$$

$$\text{Total mass NaCl} = 11.7 \text{ g}$$





Tricks



Use POMODORO technique.

earplugs for study.

Use Pen Technique.

If it is your good day turn it into great one and if it is your bad day try to minimise the losses.



Homework from modules



Preamble $\rightarrow$ Q 37 to Q 43

Prabal $\rightarrow$ Q 9

PYQ $\rightarrow$ Q 10, 11, 45, 46, 48, 49, 50, 51, 53



Magarmach Practice Questions (MPQ)



Question



What is the number of moles of O-atom in 126 amu of HNO_3 ?

A 2

C 6

B $\frac{2}{N_A}$

D $\frac{6}{N_A}$

Question



What is the charge of 96 amu of S^{2-} ?

- A** 2 C
- B** 3.2×10^{-19} C
- C** 9.6×10^{-19} C
- D** 6 C

Question



How many kg of CaCO_3 [Mol. wt. = 100 g mole^{-1}] is needed to produce 336 kg of CaO [Mol. wt. = 56 g mol^{-1}] if % yield of the reaction given is 60%.



A 10^3

B 10^2

C 900

D 800

Question



Diborane, B_2H_6 , can be prepared by the reaction;



If this reaction has a 70 percent yield, how many moles of $NaBH_4$ should be used with excess BF_3 in order to obtain 0.200 mol of B_2H_6 ?

- A** 0.200 mol
- B** 0.210 mol
- C** 0.300 mol
- D** 0.429 mol

Question



In a certain operation 358 g of TiCl_4 is reacted with 96 g of Mg. Calculate % yield of Ti if 32 g of Ti is actually obtained [At. wt. Ti = 48, Mg = 24].



A 35.46%

B 66.6%

C 100%

D 60%

Thank

You